

Roll No.....

Odd End Semester Examination December- 2024

Name of the Program: B.Tech (ECE)

Name of the Course: Network Analysis and Synthesis

Semester: III

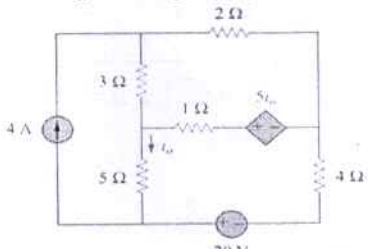
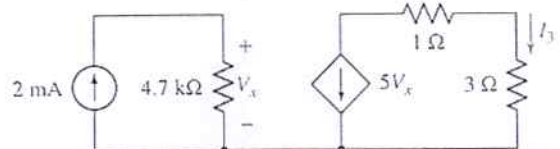
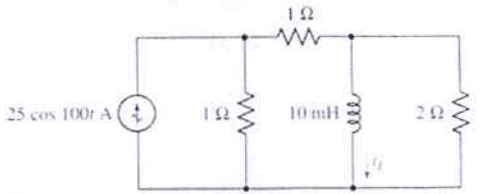
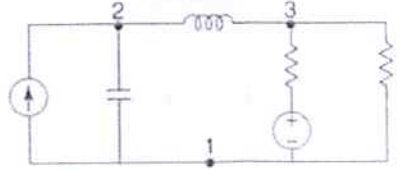
Course Code: TEC 303

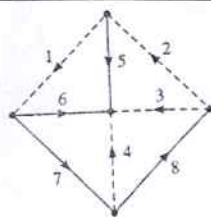
Time: 3 Hours

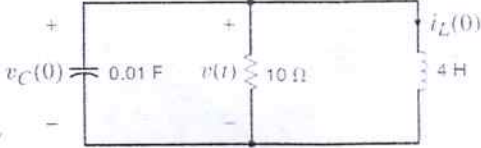
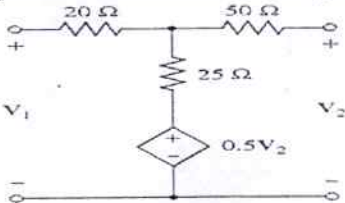
Maximum Marks: 100

Note:

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main question
- Total marks in each main question are twenty.
- Each question carries 10 marks

Q1	(20 marks)	CO-01
(a)	Find i_0 in the circuit shown in the figure using superposition. 	
(b)	Determine the current labeled I_3 in the circuit shown below. 	
(c)	Assuming there are no longer any transients present, determine the current labeled i_L in the circuit given below. 	
Q2	(20 marks)	CO-02
(a)	Draw the directed graph of the circuit shown in figure below. Write the incidence matrix of the graph and determine the maximum number of possible trees of the graph. 	
(b)	Differentiate between cut set matrix and fundamental cut set matrix with the help of an example.	
(c)	Develop Tie set matrix and determine branch currents in term of loop current.	



Q3	(20 marks)	CO-03
(a)	Derive equation for transient response in series RLC circuit and hence deduce the condition for overdamped and underdamped case.	
(b)	Determine the current $i_L(t)$ if the initial conditions are $i_L(0)=2A$ $v_C(0)=5V$. 	
(c)	A parallel resonant circuit is composed of the elements $R = 8 \text{ k}\Omega$, $L = 50 \text{ mH}$, and $C = 80 \text{ nF}$. Compute: (a) Resonant frequency (ω_0) (b) Quality Factor (Q_0) (c) Bandwidth (d) α (e) β	
Q4	(20 marks)	CO-04
(a)	Derive the expressions for the condition of reciprocity and symmetry for T parameters	
(b)	If h for some particular two-port is given by $h = \begin{bmatrix} 1 \text{ k}\Omega & -2 \\ 4 & 0.02 \text{ S} \end{bmatrix}$ calculate Y parameters.	
(c)	Obtain a complete set of Z parameters to describe the two-port network. 	
Q5	(20 marks)	CO-05 & CO-06
(a)	What are the properties of Hurwitz polynomial? Apply inspection test and conformity test to check the following polynomial: $P(s) = s^4 + s^3 + 2s^2 + 4s + 1$	
(b)	State the properties of LC immittance function. Realize the given function using Foster I form. $F(s) = \frac{s(s^2 + 4)}{(s^2 + 1)(s^2 + 9)}$	
(c)	Explain the properties of positive real function. Determine whether the given function is positive real function or not. $F(s) = \frac{(s^2 + 2s + 25)}{(s + 4)}$	